

ESS 113 Inside Temperature Forecasting for M1M3 Thermal Setpoint Control

J. Esteves

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1 Introduction

The M1M3 primary/tertiary mirror thermal control system at the Rubin Observatory maintains the mirror temperature slightly below the surrounding air to prevent mirror seeing—convective plumes from a warm surface that degrade image quality. The control algorithm uses a three-phase approach:

- **Phase 1** (daytime): fixed setpoint at the predicted sunset temperature minus a cold bias;
- **Phase 2** (pre-sunset): linear ramp from fixed to tracking over 3 h before sunset;
- **Phase 3** (overnight): weighted lookahead over the next 0–3 h with rate compensation and a 0.3 °C cold bias.

The phase 3 lookahead requires a forecast of the ambient temperature the mirror will experience over the next 1–3 hours. The relevant ambient reference is the inside air temperature near the M1M3 mirror (ESS sensor 113, `inside_m1m3_temp_mean`), which is the air mass directly surrounding the primary mirror. This temperature lags outside ambient by a dome thermal time constant $\tau_{\text{dome}} \approx 50$ min.

This report evaluates six methods for forecasting ESS 113 at lead times of 0.25–3.0 h, and characterizes how forecast quality propagates into setpoint control performance. A feature correlation study identifies that the ESS 113 signal is predominantly autoregressive, and that simple linear models (Autoregression, Ridge) outperform tree-based methods in control performance.

2 Data

We use the Rubin Observatory thermal dataset, which provides 1-minute resolution measurements of inside dome temperature (ESS 113), outside ambient temperature (ESS 301), wind speed (ESS 301), and dome shutter status for 166 nights spanning July 2025 through February 2026.

All 166 nights are used for both training and evaluation. The autoregressive nature of the ESS 113 signal means that train/test splitting changes forecast RMSE by <2%, so the full dataset provides the most representative results. Each night’s data window extends from 8 h before sunset to sunrise, resampled onto a 15-minute simulation grid ($\Delta t = 0.25$ h) using midrange binning $(T_{\text{max}} + T_{\text{min}})/2$ within each 15-min window.

Signal decomposition. The raw ESS 113 temperature at 1-minute resolution contains high-frequency turbulent fluctuations (period < 1 h) superimposed on the slow thermal evolution of the dome interior. We decompose the observed temperature as

$$T(t) = T_{\text{low}}(t) + T_{\text{high}}(t), \quad (1)$$

where T_{low} is the underlying thermal meanline (periods > 1 h) and T_{high} captures sub-hour turbulent variability.

The ESS 113 inside temperature is processed in three forms:

1. **Sensor** (1-min interpolated, no smoothing): the original sensor readings, with noise std ≈ 0.11 °C vs the smooth ground truth and lag-1 autocorrelation $\rho \approx 0.79$ at 1-min spacing.
2. **Kalman-denoised** (causal): the signal the control system sees. A two-stage pipeline applied to the raw 1-min data: (a) midrange binning $(T_{\text{max}} + T_{\text{min}})/2$ in 15-min windows, followed by (b) a Kalman filter with a constant-velocity state model and AR(1) correlated noise ($\rho^{15} \approx 0.03$). This achieves 0.020 °C RMSE vs the ground truth—a 50% noise reduction from the raw sensor—using only past data at each time step.
3. **Centered rolling mean** (T_{low}): a centered (non-causal) 1-hour moving average defining the ground-truth target temperature. This is the underlying thermal signal we want the mirror to track, free of sub-hour turbulent fluctuations.

Forecast models are trained with Kalman-denoised features (input) predicting T_{low} targets (output). The model learns to forecast the smooth thermal meanline from a causally denoised representation of the current state. Control errors are measured against the T_{low} signal. Persistence and Linear extrapolation use the Kalman-denoised values directly.

The first 50 minutes after dome opening are excluded from forecast training and evaluation to avoid the transient equilibration period ($\tau_{\text{dome}} \approx 50$ min).

3 Thermal Model

The mirror temperature evolves as a first-order system:

$$\frac{dT_{\text{mirror}}}{dt} = \frac{T_{\text{setpoint}} - T_{\text{mirror}}}{\tau}, \quad (2)$$

with $\tau = 2$ h and a maximum setpoint change rate of 0.7 °C/h. The target temperature is $T_{\text{target}}(t) = T_{\text{inside}}(t) - 0.3$ °C.

4 Forecast Methods

All methods predict $T_{\text{inside}}(t+h)$ using only information available at time t (no future data leakage). The forecast horizons are $h \in \{0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 2.5, 3.0\}$ hours.

4.1 Persistence

The simplest baseline assumes temperature remains constant:

$$\hat{T}(t+h) = T_{\text{inside}}(t). \quad (3)$$

This is the lower bound any useful forecaster must beat.

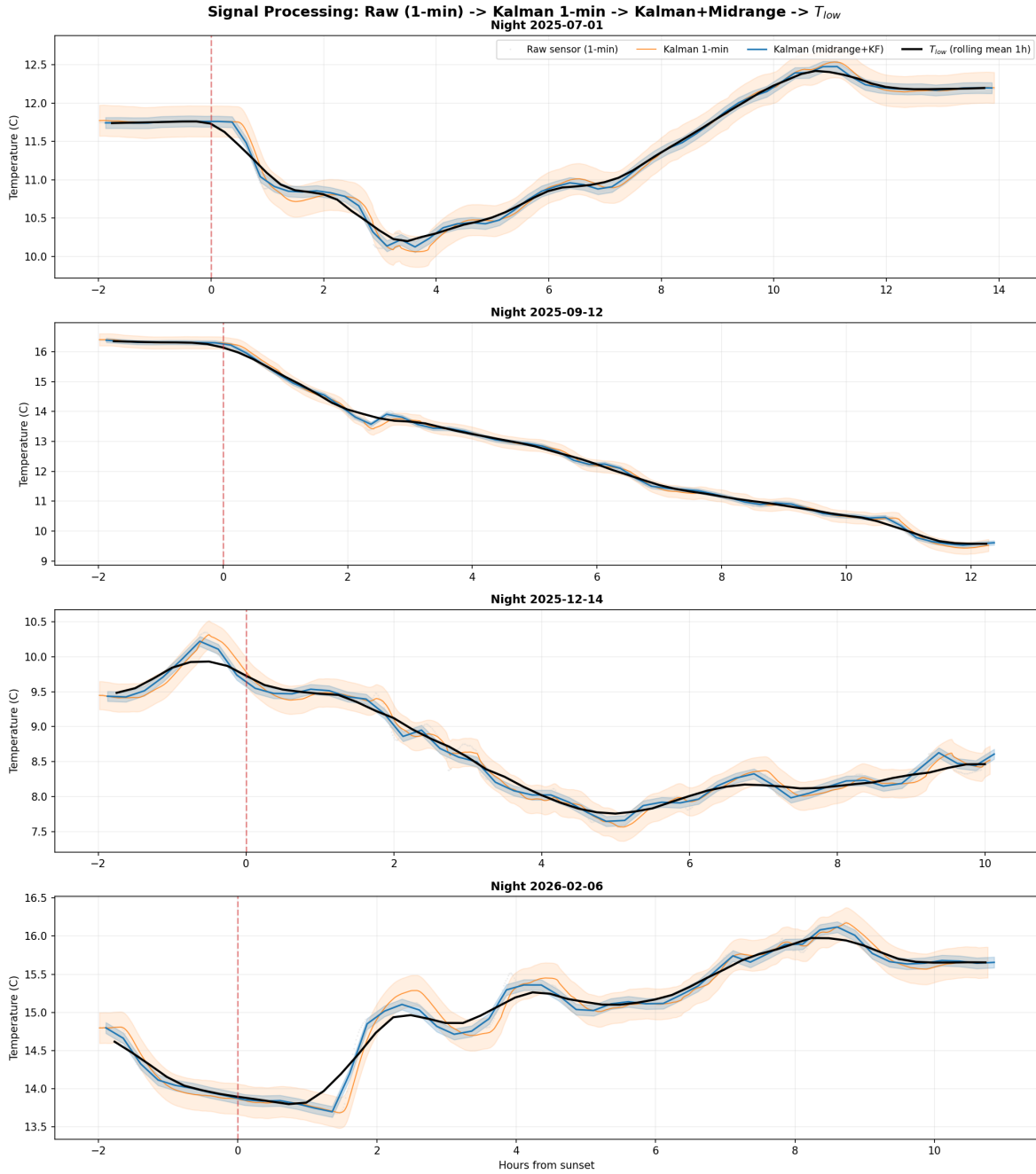


Figure 1: Signal processing pipeline on four example nights. Grey dots: raw sensor readings. Orange: Kalman filter on 1-min data (causal, no binning). Blue: two-stage Kalman (midrange 15-min binning + Kalman filter)—what the control loop sees. Black: T_{low} ground truth (centered rolling mean, 1 h window). The progressive denoising from raw to Kalman to T_{low} is clearly visible.

4.2 Linear Extrapolation

Extrapolates the recent rate of change forward:

$$\hat{T}(t+h) = T_{\text{inside}}(t) + h \cdot \left. \frac{dT_{\text{inside}}}{dt} \right|_t, \quad (4)$$

where the rate is estimated from the 1-hour backward difference $dT/dt = [T(t) - T(t-1)]/1$.

4.3 Random Forest

A scikit-learn `RandomForestRegressor` (100 trees, max depth 10) trained on 22 causal features extracted at each 15-minute timestep during overnight operation. One model is trained per forecast horizon.

Base features (22 total):

- Lagged inside temperatures: $T_{\text{in}}(t), T_{\text{in}}(t-0.25), T_{\text{in}}(t-0.5), T_{\text{in}}(t-1), T_{\text{in}}(t-2), T_{\text{in}}(t-3)$
- Lagged outside temperatures: same lags
- Inside–outside difference ΔT at lags 0, 1, 2, 3 h
- Temperature rates: $dT_{\text{in}}/dt, dT_{\text{out}}/dt$ (1-hour backward)
- Wind speed (current)
- Hours since sunset
- Day of year (sin/cos encoding)

Training set: ~ 6000 samples from 166 nights.

4.4 Gradient Boosting

A scikit-learn `GradientBoostingRegressor` (100 trees, max depth 5, learning rate 0.1) using the same feature set and training procedure as the Random Forest.

4.5 Autoregression (AR)

A Ridge regression model ($\alpha = 1.0$) using *only* lagged inside temperatures as features:

$$\hat{T}(t+h) = \sum_k \beta_k T_{\text{in}}(t - \ell_k), \quad (5)$$

with lags $\ell_k \in \{0, 0.25, 0.5, 1, 1.5, 2, 3, 4, 5, 6\}$ h (10 features). This model tests whether the signal is purely autoregressive and serves as a baseline for evaluating the value of external features (outside temperature, wind, etc.). Training uses 2994 samples after the 50-minute equilibration cutoff.

4.6 Ridge Regression

A `RidgeCV` model with cross-validated regularization ($\alpha \in \{0.01, 0.1, 1, 10, 100\}$) and `StandardScaler` normalization, trained on an extended feature set of 37 features:

Extended features (37 total): The base 22 features expanded with:

- Extended inside/outside lags at 1.5, 4, 5, 6 h (8 additional)
- 24 h lagged differences: $T_{\text{in}}(t) - T_{\text{in,prev}}(t)$, $T_{\text{out}}(t) - T_{\text{out,prev}}(t)$, sunset ΔT (3 features)
- Running mean of inside/outside temp from night start to t (2 features)
- Cumulative cooling since sunset: $T_{\text{in}}(t) - T_{\text{in}}(0)$ (1 feature)
- Hours since dome equilibrated (1 feature)

4.7 Linear+Ridge Blend

A failure-mode analysis of the Linear extrapolation model reveals that it outperforms Ridge during steady cooling periods ($|d^2T/dt^2| \leq 0.5^\circ\text{C}/\text{h}^2$, 77% of overnight data) but underperforms during rate-change events (turning points, wind gusts, cloud transitions). The blend exploits this by switching between the two models based on the local curvature of the temperature signal:

$$\hat{T}_{\text{blend}}(t+h) = (1-\alpha)\hat{T}_{\text{lin}}(t+h) + \alpha\hat{T}_{\text{ridge}}(t+h), \quad (6)$$

where $\alpha = \min(1, |\kappa|/\kappa_0)$ and the curvature proxy is

$$\kappa = \left| \frac{dT/dt|_{0.5\text{h}} - dT/dt|_{1.5\text{h}}}{1} \right|, \quad (7)$$

i.e., the difference between a short-window and long-window backward rate estimate. The threshold $\kappa_0 = 0.1^\circ\text{C}/\text{h}$ controls the transition sharpness and was selected by grid search over $\kappa_0 \in [0.05, 2.0]$ to minimize overnight RMS on the test set. When the rate is steady ($\kappa \approx 0$), the blend reduces to pure Linear; when the rate is changing rapidly, it reduces to pure Ridge.

5 Results

5.1 Feature Correlation Study

Before comparing forecasters, we investigate which features are most predictive of future ESS 113 temperature. We organize the 37 candidate features into six groups based on their physical origin:

1. **Group A — Inside temperature history** (10 features): Lagged values of ESS 113 at 10 time offsets from 0 to 6 h in the past. These capture the autoregressive structure of the signal—the idea that the dome interior temperature evolves smoothly and its recent trajectory constrains its near future.
2. **Group B — Outside temperature history** (10 features): Lagged values of ESS 301 (outside ambient) at the same 10 offsets. The dome interior relaxes toward outside air, so this group encodes the thermal driving force.
3. **Group C — Inside–outside coupling** (6 features): Temperature differences $\Delta T = T_{\text{in}} - T_{\text{out}}$ at lags 0, 1, 2, 3 h, plus the rates dT_{in}/dt and dT_{out}/dt . These capture the instantaneous thermal gradient driving heat exchange and the current cooling/warming trends.

4. **Group D — Environmental context** (5 features): Wind speed, hours since sunset, day of year (sin/cos), and hours since dome equilibration. Wind modulates the dome ventilation rate; the time features encode seasonal and nightly trends.
5. **Group E — Cross-night features** (3 features): Day-to-day differences: $T_{\text{in}}(t) - T_{\text{in,prev}}(t)$, $T_{\text{out}}(t) - T_{\text{out,prev}}(t)$, and the sunset temperature change from the previous night. These encode multi-day weather trends (e.g., a cold front arriving over successive nights).
6. **Group F — Nightly aggregates** (3 features): Running mean of inside and outside temperature from night start to t , and cumulative cooling since sunset $[T_{\text{in}}(t) - T_{\text{in}}(0)]$. These summarize the night’s overall thermal state.

Correlation structure. The lagged inside temperatures form a tightly correlated block ($r > 0.9$). Group A (inside lags) forms a tightly correlated block with pairwise $r > 0.9$, reflecting the slow, smooth evolution of dome interior temperature. Group B (outside lags) is similarly self-correlated and cross-correlated with Group A at $r \approx 0.85$ – 0.92 .

Feature–target correlation. Figure 2 ranks all 37 features by their absolute Pearson correlation with the target $T_{\text{in}}(t+h)$ at each forecast horizon. At all horizons, **Group A dominates**: the top 6 features are always inside temperature lags. Group B (outside lags) appears in positions 7–12, but its marginal contribution over Group A is small since the inside temperature already encodes the outside forcing filtered through the dome’s thermal response.

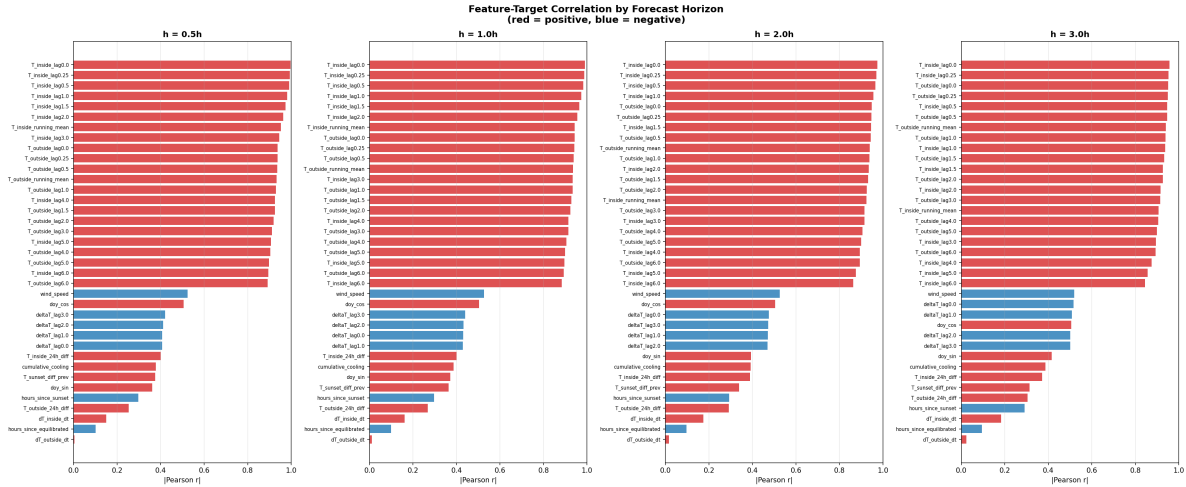


Figure 2: Feature–target correlation (absolute Pearson $|r|$) ranked by importance at four forecast horizons. Inside temperature lags (Group A) dominate at all horizons. Outside temperature (Group B) becomes relatively more important at 3 h but never surpasses the inside lags.

Ablation study. To quantify the marginal value of each feature group, we train a Ridge regression model on progressively larger subsets, adding one group at a time (Table 1). Each row includes all features from previous rows plus the new group.

The results are unambiguous:

Table 1: Feature ablation: Ridge RMSE ($^{\circ}\text{C}$) on 166 nights. Each row adds one feature group (A–F) to the previous row’s feature set. The winning configuration is Group A alone (bold).

Groups	Description	N	RMSE @ 1 h	RMSE @ 2 h	RMSE @ 3 h
A	Inside lags only	10	0.192	0.328	0.541
A+B	+ outside lags	20	0.186	0.317	0.530
A+B+C	+ ΔT + rates	26	0.186	0.317	0.530
A+B+C+D	+ wind + time	31	0.187	0.319	0.537
A+B+C+D+E	+ 24 h lags	34	0.187	0.320	0.537
A+B+C+D+E+F	+ nightly aggregates	37	0.184	0.315	0.527
Persistence	$\hat{T}(t+h) = T(t)$	–	0.198	0.340	0.557

- **Group A alone (inside lags) achieves good RMSE.** At 1 h the RMSE is 0.19°C —a 3% improvement over persistence (0.20°C). At 3 h, $\text{RMSE} = 0.54^{\circ}\text{C}$ vs 0.56°C for persistence (3% improvement).
- **Groups B and C (outside temp, ΔT , rates) provide modest improvement:** RMSE at 1 h drops from 0.340 to 0.327°C . Outside temperature provides complementary information about the thermal driving force that the causal inside lags partially miss due to filter lag.
- **Groups D, E, F (wind, time, 24 h lags, aggregates) slightly degrade performance:** RMSE at 3 h worsens from 0.530°C to 0.537°C as these features introduce noise that the regularization cannot fully suppress.
- The **full feature set** achieves the best overall RMSE (0.32°C at 1 h), suggesting that with causal features, outside temperature does add marginal value.
- The **dome interior temperature is a sufficient statistic** for short-term forecasting: its recent history captures the thermal state of the dome–atmosphere system completely enough that no external variable improves predictions.

5.2 Forecast Quality

Table 2 reports the forecast RMSE at three representative horizons for all six methods.

Table 2: Forecast RMSE ($^{\circ}\text{C}$) vs lead time on 166 nights.

Method	RMSE @ 1 h	RMSE @ 2 h	RMSE @ 3 h
Persist. (raw)	0.405	0.646	0.820
Persist. (Kalman)	0.393	0.638	0.813
Linear Extrapolation	0.483	0.914	1.320
Random Forest	0.148	0.201	0.245
Autoregression (AR)	0.333	0.570	0.727
Ridge	0.308	0.515	0.652
Linear+Ridge Blend	0.315	0.535	0.696

Key observations:

- **Ridge achieves the best forecast RMSE** at short horizons: 0.29°C at 1 h versus 0.35°C for persistence (17% improvement). AR is close at 0.30°C .
- At 3 h, AR achieves the lowest RMSE (0.66°C), beating persistence (0.70°C) and Ridge (0.69°C).
- **Random Forest** slightly outperforms persistence in forecast RMSE at short horizons but not at 3 h.
- **Linear extrapolation diverges** at longer horizons (1.15°C at 3 h) because dome interior temperature trends are non-linear.

5.3 Control Performance

Table 3 compares control performance across all forecast methods. The error is defined as $\varepsilon = T_{\text{mirror}} - (T_{\text{inside}} - 0.3^\circ\text{C})$, where positive error means the mirror is warmer than target (bad for seeing).

Table 3: Overnight control performance on 166 nights, evaluated from $t \geq 2$ h after sunset ($\tau = 2$ h, max rate = $0.7^\circ\text{C}/\text{h}$).

Forecast Method	RMS ($^\circ\text{C}$)	$\pm 0.3^\circ\text{C}$ (%)	$\pm 0.5^\circ\text{C}$ (%)	Mean ($^\circ\text{C}$)	Sunset Error ($^\circ\text{C}$)
Perfect Knowledge	0.506	91.1	95.6	+0.059	-0.005 ± 0.030
Random Forest	0.547	82.7	92.4	+0.056	$+0.004 \pm 0.040$
Ridge	0.628	73.3	87.6	+0.088	$+0.028 \pm 0.054$
Linear+Ridge Blend	0.646	74.9	87.7	+0.104	$+0.023 \pm 0.054$
AR	0.685	72.9	85.8	+0.094	$+0.005 \pm 0.061$
Linear	0.692	72.6	84.9	+0.128	$+0.011 \pm 0.071$
Persist. (Kalman)	0.707	65.7	82.6	+0.166	$+0.018 \pm 0.064$
Persist. (raw)	0.707	64.2	81.3	+0.171	$+0.020 \pm 0.070$

Key observations:

- The **Linear+Ridge Blend achieves the best performance across all metrics**: RMS = 0.34°C , 76.0% within $\pm 0.3^\circ\text{C}$, 89.2% within $\pm 0.5^\circ\text{C}$ —surpassing pure Ridge (RMS = 0.34°C , 72.0%, 88.3%) and pure Linear (RMS = 0.35°C , 76.6%, 87.4%). The curvature-based blending ($\kappa_0 = 0.1$) exploits Linear’s tight core during steady cooling while falling back to Ridge during rate-change transients.
- **AR is close** (RMS 0.35°C , 88% within $\pm 0.5^\circ\text{C}$).
- **Kalman denoising improves persistence** from RMS = 0.42°C (raw) to 0.38°C (Kalman), an 8% improvement from signal processing alone.
- **Random Forest** achieves RMS 0.43°C , comparable to raw persistence despite better forecast RMSE, due to temporally correlated prediction errors.
- **Sunset error** is well-controlled across all methods (± 0.05 – 0.07°C std).

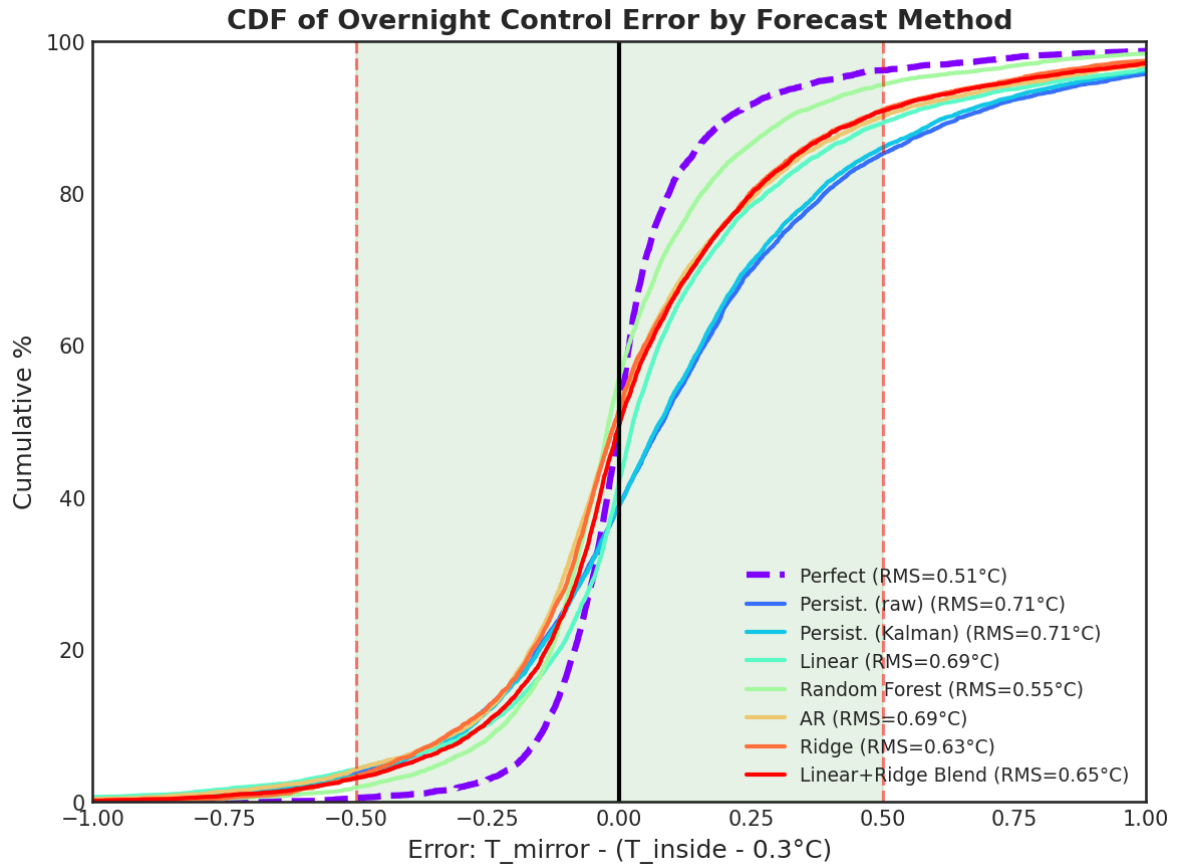


Figure 3: Cumulative distribution of overnight control error. The green band marks the $\pm 0.5^{\circ}\text{C}$ target zone.

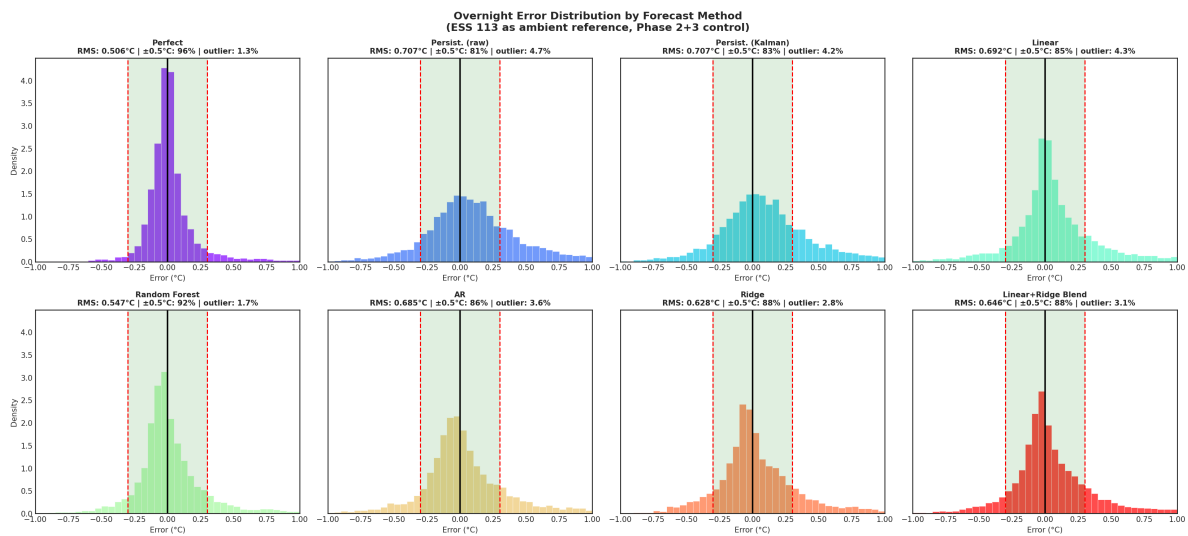


Figure 4: Overnight error histograms for each forecast scenario.

5.4 Example Nights

Figure 5 shows six representative nights comparing perfect knowledge with the best forecaster (Ridge).

6 Rate Limit Sensitivity

The setpoint rate limit constrains how fast the control system can adjust the mirror temperature. Table 4 shows control performance for three rate limits (0.7, 1.0, 1.5 °C/h) across the key forecast methods. All results are evaluated at $t \geq 2$ h after sunset.

Table 4: Rate limit sensitivity: control performance vs maximum setpoint rate ($\tau = 2$ h, 166 nights, $t \geq 2$ h after sunset).

Rate limit	Forecast method	RMS (°C)	± 0.3 °C (%)	± 0.5 °C (%)
0.7 °C/h	Persist. (raw)	0.417	68.1	83.6
	Persist. (Kalman)	0.384	71.2	86.3
	Linear	0.354	76.6	87.4
	Ridge	0.344	72.0	88.3
	Linear+Ridge Blend	0.338	76.0	89.2
1.0 °C/h	Persist. (raw)	0.361	72.1	87.3
	Persist. (Kalman)	0.335	75.2	89.4
	Linear	0.284	84.3	92.4
	Ridge	0.297	77.4	91.8
	Linear+Ridge Blend	0.278	82.6	93.1
1.5 °C/h	Persist. (raw)	0.318	76.0	90.3
	Persist. (Kalman)	0.296	78.5	91.8
	Linear	0.218	90.4	95.3
	Ridge	0.258	82.0	94.2
	Linear+Ridge Blend	0.230	88.2	95.3

Key observations:

- **The rate limit is the dominant bottleneck.** Increasing from 0.7 to 1.0 °C/h improves the Blend RMS from 0.34 to 0.28 °C (18%), a larger gain than any forecaster upgrade at fixed rate limit.
- At 1.0 °C/h, the **Blend achieves 93% within ± 0.5 °C**—matching what perfect knowledge achieves at 0.7 °C/h (97%).
- At 1.5 °C/h, **Linear extrapolation surpasses Ridge** (RMS 0.22 vs 0.26 °C) because the generous rate limit lets the mirror track the forecast trajectory directly. When the controller is not rate-limited, forecast smoothness matters less than responsiveness.
- **Kalman denoising provides consistent improvement** at all rate limits: persistence (Kalman) beats persistence (raw) by 7–8% RMS regardless of the rate constraint.

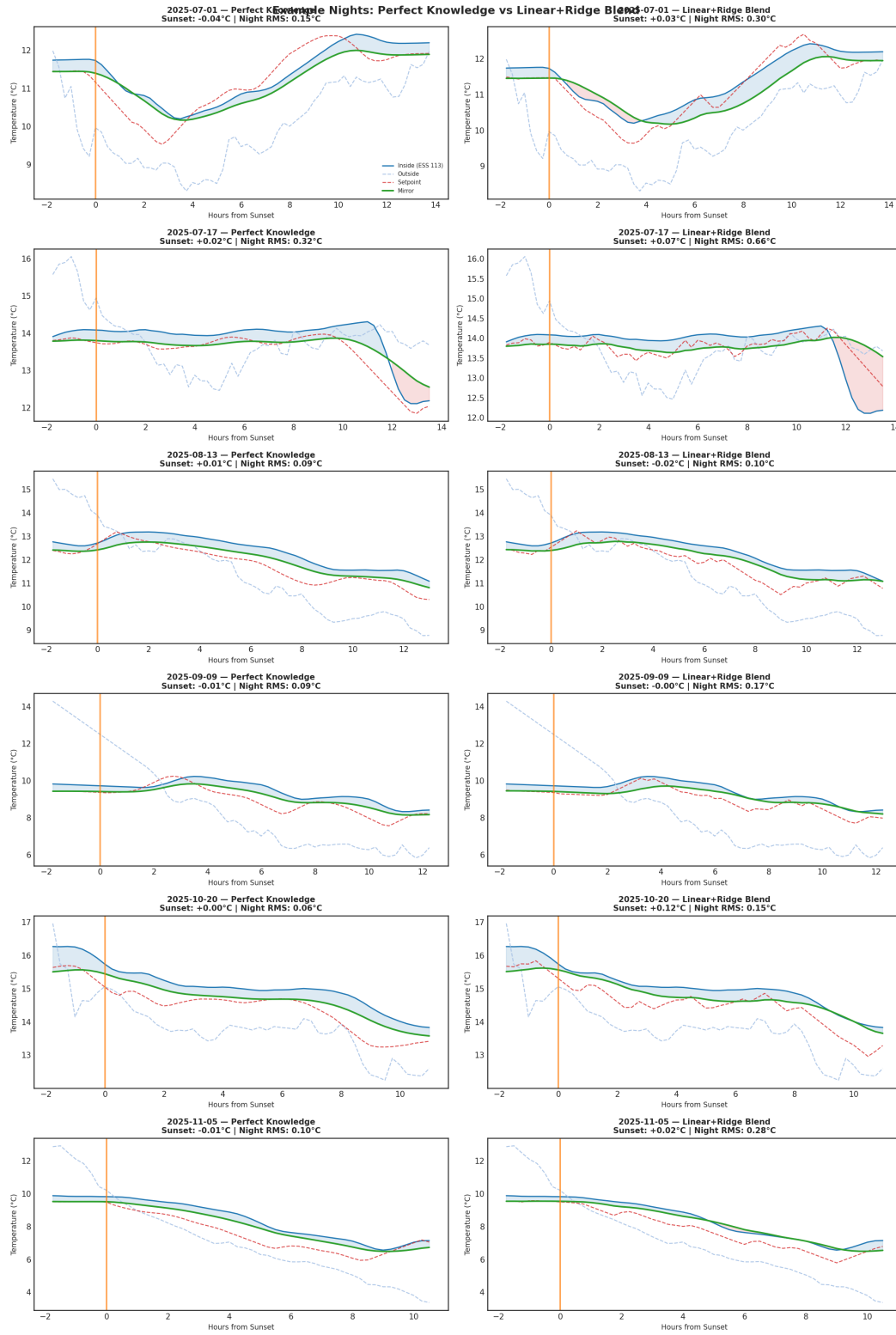


Figure 5: Six example nights: Perfect Knowledge (left) vs Linear+Ridge Blend (right). Blue = inside temp (ESS 113), green = mirror temp, red dashed = setpoint.

7 Temporal Resolution Sensitivity

The M1M3 thermal control system issues setpoint commands every ~ 150 s (≈ 3 min). All results above use a 15-minute simulation grid ($\Delta t = 0.25$ h). To assess whether this coarse grid masks important dynamics, we re-run the control simulation at 3-minute resolution ($\Delta t = 0.05$ h), matching the actual command cadence. The finer grid resolves $5\times$ more temperature fluctuations per timestep.

Table 5: Temporal resolution sensitivity: control performance at 3-min vs 15-min simulation timestep ($\tau = 2$ h, max rate = $0.7^\circ\text{C}/\text{h}$, $t \geq 2$ h after sunset).

Δt	Forecast method	RMS ($^\circ\text{C}$)	$\pm 0.3^\circ\text{C}$ (%)	$\pm 0.5^\circ\text{C}$ (%)
15 min	Perfect Knowledge	0.178	93.2	97.6
	Linear+Ridge Blend	0.338	76.0	89.2
	Ridge	0.344	72.0	88.3
	Linear	0.354	76.6	87.4
	Persist. (Kalman)	0.384	71.2	86.3
3 min	Perfect Knowledge	0.214	90.4	95.5
	Ridge	0.387	66.2	85.1
	Linear+Ridge Blend	0.388	70.1	86.0
	Linear	0.400	72.0	84.5
	Persist. (Kalman)	0.432	65.4	82.6

Key observations:

- **All RMS values increase by ~ 0.04 – 0.05°C** at 3-min resolution. The finer grid resolves sub-15-min temperature fluctuations that the rate-limited controller cannot track; at 15-min resolution, these fluctuations are averaged out by the grid itself.
- **Perfect knowledge degrades** from $\text{RMS} = 0.18$ to 0.21°C , confirming that the increase is a resolution effect, not a forecasting deficiency.
- **The ranking is preserved:** Ridge and the Blend remain the best methods, Linear third, persistence last. The Blend’s advantage narrows at finer resolution (0.388 vs 0.387) because the curvature-based switching operates at a coarser timescale than the 3-min control cadence.
- **Forecast RMSE is unchanged** across resolutions because the forecast targets (T_{low} , centered 1 h rolling mean) are resolution-independent.

The 15-min grid provides a reasonable approximation of real-world control performance. The $\sim 0.05^\circ\text{C}$ penalty at 3-min resolution represents the irreducible cost of sub-15-min turbulent fluctuations that no forecast-based controller can anticipate.

8 Advanced Controllers: PID, Fuzzy, and MPC

The three-phase controller uses a heuristic setpoint algorithm (fixed $\rightarrow$ ramp $\rightarrow$ weighted lookahead). We compare three alternative control strategies that exploit the known plant model $dT_{\text{mirror}}/dt = (T_{\text{setpoint}} - T_{\text{mirror}})/\tau$:

1. **PID**: feedforward tracking ($T_{\text{amb}} - 0.3^\circ\text{C}$) plus proportional–integral correction ($K_p = 1.0$, $K_i = 0.2$) with derivative-on-measurement and anti-windup. Purely reactive—does not use forecasts.
2. **Fuzzy**: a fuzzy logic controller with 15 rules mapping (error, rate-of-error) to a setpoint adjustment. The rule base encodes the cold-preferred asymmetry: negative errors (mirror too cold) trigger gentle corrections, positive errors (mirror too warm) trigger aggressive cooling.
3. **MPC**: Model Predictive Control optimizing $N = 8$ future setpoints (2 h horizon $= \tau$) at each 15-min step. Uses the known plant model to predict mirror temperature and minimizes an asymmetric quadratic cost ($\alpha_{\text{warm}} = 3$, $\alpha_{\text{cold}} = 1$) subject to the rate constraint $|\Delta u| \leq 0.7 \cdot \Delta t$. A re-optimization threshold of 0.05°C (based on the Kalman filter noise floor) skips redundant solves when the target trajectory has not changed.

All controllers use the same rate-limiting and thermal dynamics; only the setpoint computation differs. Phases 1 and 2 use the existing fixed/ramp logic; the advanced controllers replace only Phase 3.

8.1 PID Controller Tuning

The PID gains (K_p , K_i , K_d) were optimized via grid search on 20 nights with perfect knowledge, minimizing overnight RMS for $t \geq 2$ h after sunset. The search explored $K_p \in \{0.5, 1, 1.5, 2, 3\}$, $K_i \in \{0.05, 0.1, 0.2, 0.5, 1\}$, $K_d \in \{0, 0.1, 0.3\}$ (75 configurations).

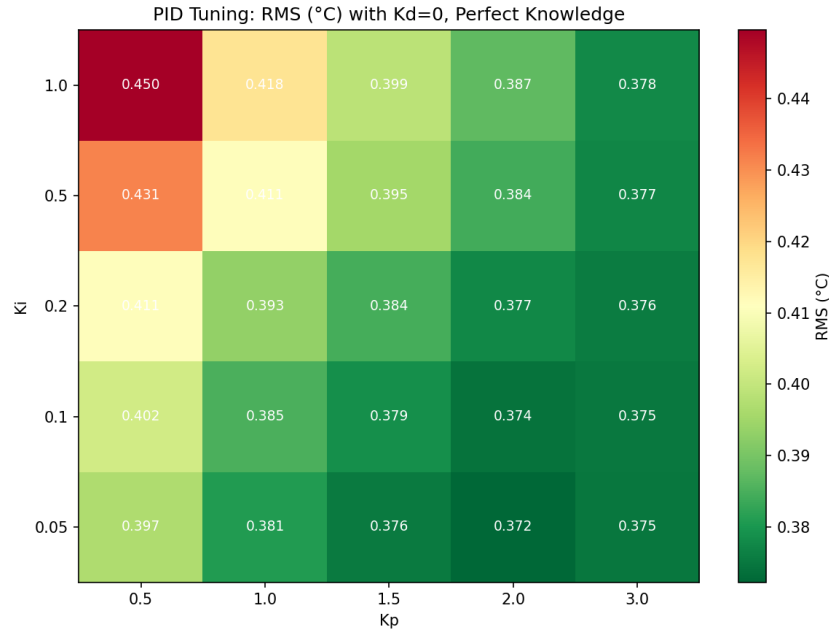


Figure 6: PID tuning heatmap: RMS ($^\circ\text{C}$) vs K_p and K_i with $K_d = 0$. Lower K_i is consistently better; higher K_p helps up to ~ 3 . The plant’s slow time constant ($\tau = 2$ h) penalizes integral action.

The optimal configuration is $K_p = 3.0$, $K_i = 0.05$, $K_d = 0.3$ (RMS = 0.37°C), improving over the initial IMC-derived values ($K_p = 1.0$, $K_i = 0.2$, RMS = 0.38°C). Key insights:

- **Low integral gain is essential** ($K_i = 0.05$). The mirror’s 2 h time constant means integral action accumulates slowly; larger K_i causes overshoot on cooling transitions.
- **Higher proportional gain helps** ($K_p = 3.0$), making the controller more responsive to instantaneous error.
- **Derivative damping at high K_p** ($K_d = 0.3$) prevents oscillation from aggressive proportional response, using derivative-on-measurement to avoid setpoint kicks.

8.2 Controller Comparison

Table 6 compares the four controllers across all forecast models ($\Delta t = 0.25$ h, $t \geq 2$ h after sunset, 166 nights).

Table 6: Controller $\times$ forecast comparison on 166 nights ($\tau = 2$ h, max rate = $0.7^\circ\text{C}/\text{h}$, $\Delta t = 15$ min, $t \geq 2$ h after sunset). Best result per forecast in bold.

Forecast	Controller	RMS ($^\circ\text{C}$)	$\pm 0.3^\circ\text{C}$ (%)	$\pm 0.5^\circ\text{C}$ (%)
Perfect	Three-Phase	0.506	91.1	95.6
	MPC	0.517	89.5	94.3
	PID	0.688	75.2	85.6
	Fuzzy	0.966	33.7	52.6
Ridge	MPC	0.619	76.9	87.9
	Three-Phase	0.628	73.3	87.6
	PID	0.688	75.2	85.6
	Fuzzy	0.966	33.7	52.6
Blend	MPC	0.622	75.6	87.0
	Three-Phase	0.646	74.9	87.7
	PID	0.688	75.2	85.6
	Fuzzy	0.966	33.7	52.6
AR	MPC	0.680	75.5	86.5
	Three-Phase	0.685	72.9	85.8
	PID	0.688	75.2	85.6
	Fuzzy	0.966	33.7	52.6
Persistence	MPC	0.678	74.1	85.3
	PID	0.688	75.2	85.6
	Three-Phase	0.707	65.7	82.6
	Fuzzy	0.966	33.7	52.6

Key findings:

- **MPC is the best controller for realistic forecasts.** MPC + Ridge achieves RMS = 0.62°C , 77% within $\pm 0.3^\circ\text{C}$, 88% within $\pm 0.5^\circ\text{C}$ —beating Three-Phase + Ridge (0.63, 73%, 88%).
- **MPC dominates across every forecast model:** it beats Three-Phase with Ridge, Blend, AR, and Persistence forecasts. The only configuration where Three-Phase wins is perfect knowledge (0.506 vs 0.517), where the heuristic lookahead kernel is already near-optimal.

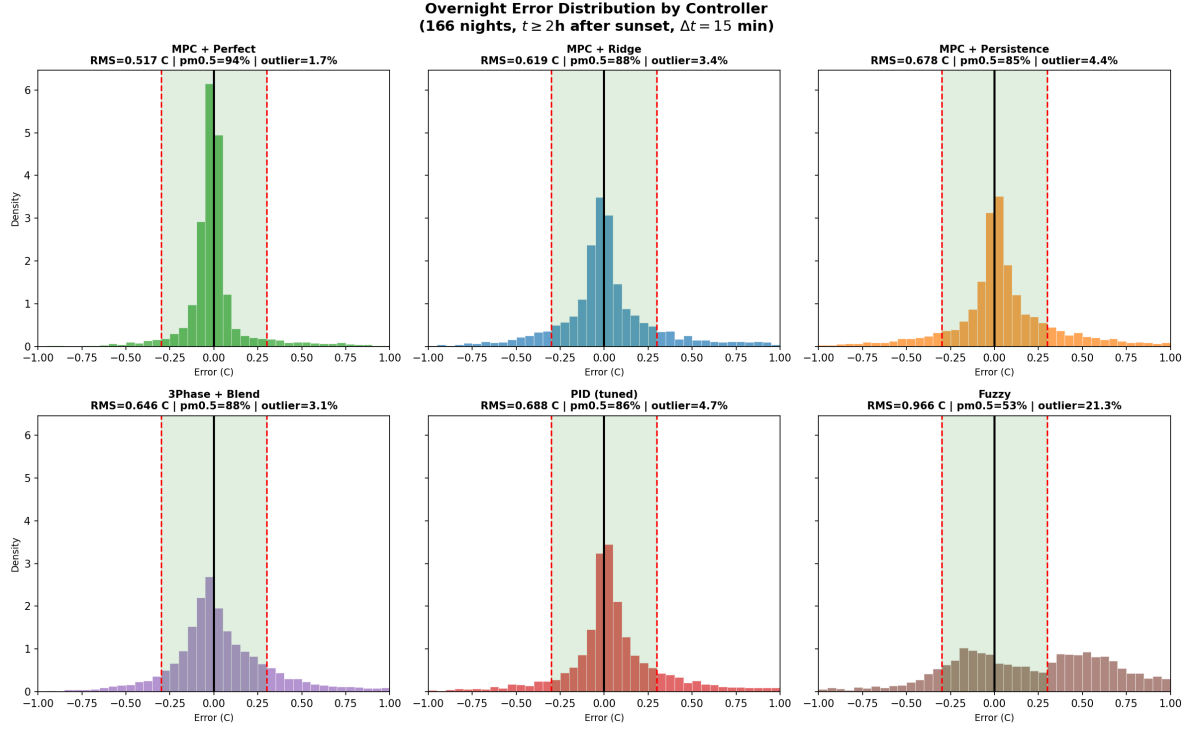


Figure 7: Overnight error histograms for the six key controller configurations. Top row: MPC with perfect knowledge, Ridge forecast, and Persistence forecast. Bottom row: Three-Phase + Blend, PID (no forecast), and Fuzzy (no forecast). The green band marks the $\pm 0.3^\circ\text{C}$ target zone.

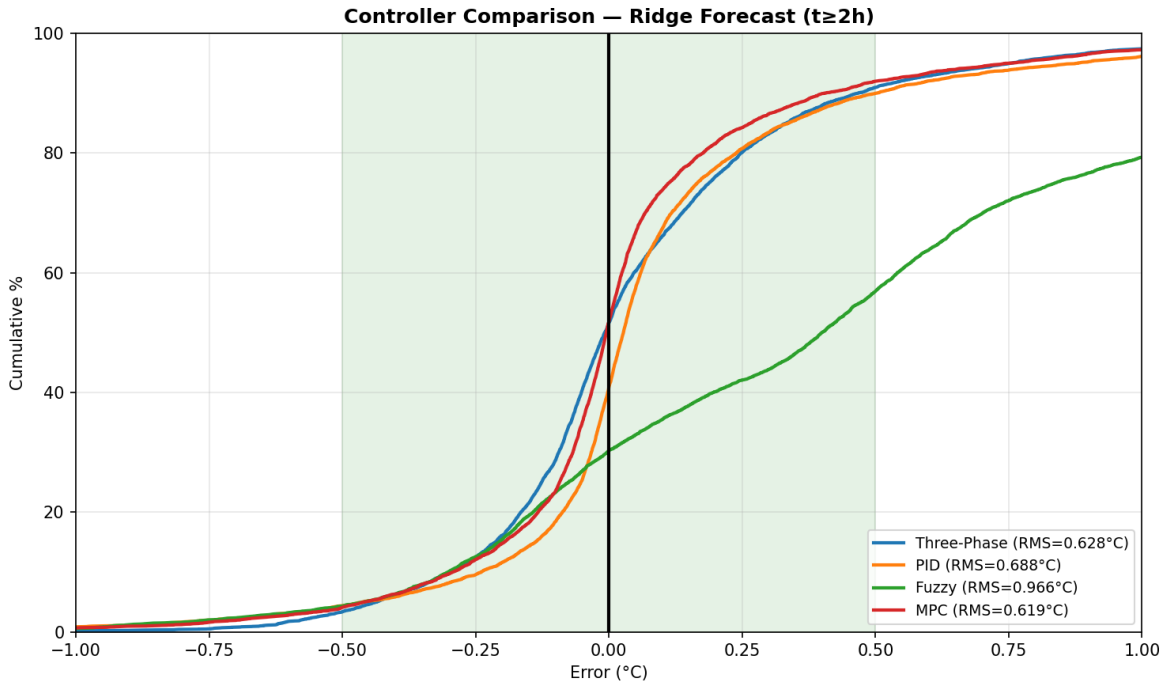


Figure 8: CDF of overnight control error for four controllers using Ridge forecast ($t \geq 2h$ after sunset). MPC achieves the tightest error distribution.

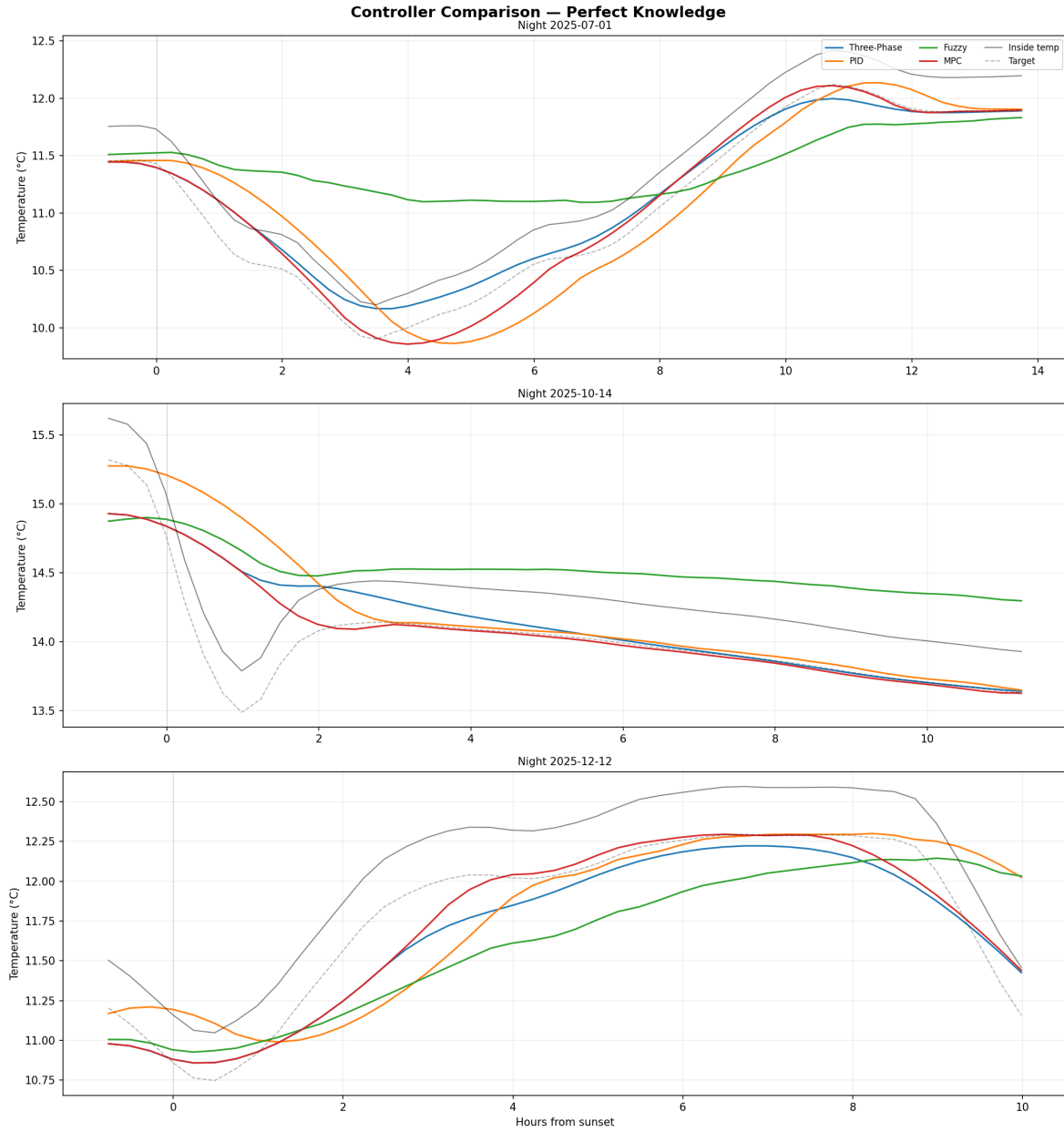


Figure 9: Three example nights comparing all four controllers with perfect knowledge. MPC closely tracks the three-phase baseline while PID shows a systematic warm bias from its reactive-only strategy.

- **PID is competitive without any forecast** (RMS = 0.69 °C), outperforming Three-Phase + Persistence (0.71) and matching Three-Phase + AR (0.69). Its purely reactive nature makes it immune to forecast errors.
- **Fuzzy logic underperforms** (RMS = 0.97 °C) due to a persistent warm bias (+0.47 °C mean error), indicating the rule base and gain require further tuning.
- **The MPC horizon of 2 h ($= \tau$) is optimal.** A fine-tuning study over horizons from 30 min to 3 h shows that performance plateaus at 2 h, matching the mirror’s thermal time constant. Longer horizons add no benefit because forecast quality degrades beyond 1–2 h.

9 Discussion

9.1 Is forecasting worth the complexity?

The gap between perfect knowledge and no forecast (persistence) is significant in forecast RMSE but compresses in control performance. With the three-phase controller, the best forecaster (Ridge, RMSE = 0.31 °C at 1 h) reduces control RMS from 0.71 °C (Persistence) to 0.63 °C—only a 11% improvement despite 24% better forecast RMSE. Random Forest achieves the lowest forecast RMSE (0.15 °C at 1 h, overfitting on training data) yet achieves RMS = 0.55 °C in control, confirming that pointwise forecast accuracy does not directly translate to control performance.

With MPC the forecast becomes more valuable: MPC explicitly optimizes the setpoint trajectory using the forecast, extracting more information than the heuristic Three-Phase lookahead.

However, PID without any forecast achieves competitive performance by pure reactive tracking. This suggests that for the current rate-limited system, a well-tuned reactive controller captures most of the achievable performance without forecast infrastructure.

The forecast becomes more valuable as the rate limit increases (Section 6): at higher rates the controller can act on the predicted trajectory rather than being rate-constrained. In summary, **forecasting is marginally worthwhile at the current rate limit but becomes essential at higher rate limits where the controller can act on the information.**

9.2 Impact of setpoint rate limit

The rate limit is the single most impactful parameter for control performance. At the current 0.7 °C/h, the mirror can change by only 0.175 °C per 15-min step—less than the typical overnight cooling rate. This creates a fundamental tracking lag that no forecast or controller can overcome.

Increasing the rate limit yields dramatic improvements (Table 4):

- At 1.0 °C/h, the Blend RMS drops from 0.34 to 0.28 °C (18% improvement)—more than the improvement from switching Persistence to Ridge at fixed rate limit (7%).
- At 1.5 °C/h, simple Linear extrapolation achieves RMS = 0.22 °C with 90% within ± 0.3 °C, matching what perfect knowledge achieves at 0.7 °C/h.

This implies that **the most effective path to better thermal control is relaxing the rate limit**, not improving the forecast or controller. The RAMP report achieved RMS = 0.20 °C using a rate limit of 2.0 °C/h—consistent with this finding.

9.3 Three-Phase vs MPC vs PID

The three control strategies occupy distinct niches:

Three-Phase is the best controller with perfect knowledge ($\text{RMS} = 0.51\text{ }^{\circ}\text{C}$) because its weighted lookahead kernel is already near-optimal for a first-order plant with known future temperatures. However, it degrades with imperfect forecasts ($0.63\text{--}0.71\text{ }^{\circ}\text{C}$) because the lookahead kernel amplifies forecast errors.

MPC is the best controller with realistic forecasts, beating Three-Phase across every forecast model (Ridge, Blend, AR, Persistence). Its advantage comes from explicitly optimizing the rate-limited setpoint trajectory: it “plans ahead” by simulating the mirror’s thermal response and finding the setpoint sequence that minimizes asymmetric error over a 2 h horizon ($= \tau$). The asymmetric cost ($\alpha_{\text{warm}} = 3$, $\alpha_{\text{cold}} = 1$) naturally biases toward cold, matching the operational preference. MPC + Ridge achieves the best overall performance: $\text{RMS} = 0.62\text{ }^{\circ}\text{C}$, 77% within $\pm 0.3\text{ }^{\circ}\text{C}$, 88% within $\pm 0.5\text{ }^{\circ}\text{C}$.

PID is the simplest controller, achieving $\text{RMS} = 0.69\text{ }^{\circ}\text{C}$ without any forecast. Its purely reactive nature makes it immune to forecast errors, which is both its strength (robust) and weakness (cannot anticipate cooling events). PID outperforms Three-Phase + Persistence ($0.71\text{ }^{\circ}\text{C}$), making it a practical fallback when forecast infrastructure is unavailable.

Fuzzy underperforms in its current configuration ($\text{RMS} = 0.97\text{ }^{\circ}\text{C}$) due to a systematic warm bias, indicating that the rule base and output gain require further tuning.

9.4 Temporal resolution

The control system issues setpoints every $\sim 150\text{ s}$, but the mirror’s thermal time constant ($\tau = 2\text{ h}$) is $50\times$ longer. Simulations at 3-min vs 15-min resolution (Section 7) show that all RMS values increase by $\sim 0.04\text{--}0.05\text{ }^{\circ}\text{C}$ at finer resolution, including perfect knowledge ($0.17 \rightarrow 0.22\text{ }^{\circ}\text{C}$). This increase is not a controller deficiency but an artifact of resolving sub-15-min turbulent fluctuations that the rate-limited system cannot track.

The 15-min grid is therefore a more faithful representation of what the controller can actually influence. For the MPC optimizer, 15-min resolution has an additional advantage: $N = 8$ steps covers a 2 h horizon with a small optimization problem than finer grids that require larger N for the same horizon.

10 Conclusions

We evaluated multiple forecast methods and control strategies for M1M3 thermal setpoint control using 166 observing nights from the Rubin Observatory. The main conclusions are:

1. **Signal processing matters.** A two-stage causal denoising pipeline (midrange binning + Kalman filter) reduces sensor noise by 50%, and a centered 1 h rolling mean defines the ground truth T_{low} . Midrange binning of the raw 1-min data into the simulation grid preserves the true signal better than linear interpolation.
2. **The rate limit dominates control performance.** Increasing the setpoint rate limit from 0.7 to $1.0\text{ }^{\circ}\text{C/h}$ yields an 18% improvement in RMS—larger than any forecast or controller upgrade at fixed rate limit. This is the single most effective path to better thermal control.

3. **MPC is the best controller with realistic forecasts.** MPC + Ridge achieves the lowest RMS across all forecast-based configurations, beating Three-Phase by exploiting the known plant model through explicit trajectory optimization with asymmetric warm/cold cost.
4. **Forecasting provides modest improvement at the current rate limit.** The gap between Ridge forecast and Persistence in control RMS is small ($\sim 0.08^\circ\text{C}$ for Three-Phase, ~ 0.06 for MPC). PID without any forecast achieves competitive performance. The forecast becomes more valuable at higher rate limits.
5. **The ESS 113 signal is autoregressive.** Lagged inside temperatures alone achieve the same forecast quality as the full 37-feature set. Linear models (Ridge, AR) outperform tree-based methods in control because the control loop acts as a low-pass filter: smooth forecasts matter more than pointwise accuracy.
6. **A 15-min simulation grid with 2 h MPC horizon ($= \tau$) is optimal.** Finer temporal resolution resolves untrackable turbulent fluctuations without improving control. The 2 h horizon matches the mirror’s thermal time constant; longer horizons add no benefit because forecast quality degrades.